

PITTSBURG-BUTLER, PA, USA

WMO: 725124

Lat: 40.777N

Lon: 79.950W

Elev: 1247

StdP: 14.05

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 64705

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 2.6	(c) 8.0	(d) -6.4	(e) 4.1	(f) 5.3	(g) -0.9	(h) 5.5	(i) 10.5	(j) 22.1	(k) 31.6	(l) 19.5	(m) 29.5	(n) 5.0	(o) 260	(p) 0.461

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.0	88.1	72.3	84.5	70.9	82.3	69.4	74.9	83.9	73.3	81.7	71.8	79.5	7.1	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.2	125.0	80.1	70.5	117.5	77.7	69.8	114.8	77.0	39.2	84.3	37.8	82.0	36.3	78.9	80.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 18.2	(b) 16.0	(c) 13.4	(e) -3.7	(f) 91.1	(g) 7.4	(h) 3.0	(i) -9.0	(j) 93.2	(k) -13.3	(l) 95.0	(m) -17.5	(n) 96.6	(o) -22.8	(p) 98.8		
(a) 18.2	(b) 16.0	(c) 13.4	(e) -4.2	(f) 77.4	(g) 6.7	(h) 1.8	(i) -9.1	(j) 78.7	(k) -13.0	(l) 79.8	(m) -16.8	(n) 80.8	(o) -21.7	(p) 82.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.1	27.1	29.4	37.9	49.7	59.2	67.5	71.2	69.6	63.0	51.9	41.3	32.6
	DBStd	17.85	12.26	11.53	11.17	9.54	8.21	6.21	5.07	5.11	6.95	8.35	9.33	10.01
	HDD50	2756	716	583	408	121	15	0	0	0	1	77	290	545
	HDD65	6021	1176	997	842	466	218	46	9	17	119	416	711	1004
	CDD50	2807	6	6	31	112	301	525	657	608	391	135	29	6
	CDD65	596	0	0	1	7	39	121	201	160	58	9	0	0
	CDH74	4768	0	0	8	78	385	989	1611	1185	454	58	0	0
	CDH80	1243	0	0	0	9	74	260	473	309	109	9	0	0
	WSAvg	5.6	7.1	6.7	6.8	6.8	5.5	4.6	4.1	3.8	4.3	5.3	6.1	6.6
	PrecAvg	40.8	3.4	2.6	3.3	3.5	3.7	4.1	4.1	3.7	3.7	2.8	3.2	3.3
(15) Precipitation	PrecMax	54.7	7.1	5.4	5.5	6.1	5.7	6.7	9.6	7.5	8.9	6.3	5.4	7.2
	PrecMin	29.5	2.2	1.2	1.7	1.8	1.4	1.1	1.4	1.6	1.7	0.8	0.9	1.5
	PrecStd	5.7	1.2	0.9	1.1	1.0	1.3	1.5	1.9	1.6	1.8	1.1	1.3	1.3
	PrecStd	5.7	1.2	0.9	1.1	1.0	1.3	1.5	1.9	1.6	1.8	1.1	1.3	1.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.9	64.1	73.3	81.3	85.9	89.8	90.9	90.3	88.1	80.8	70.1	63.0
		MCWB	56.0	52.9	58.6	62.4	69.6	72.9	74.7	72.0	71.0	67.8	58.8	55.9
	2%	DB	55.9	57.4	67.6	75.3	81.8	86.0	88.1	86.5	82.7	73.5	64.2	57.0
		MCWB	51.7	49.7	54.5	58.9	66.4	71.2	73.2	71.4	69.1	62.1	55.0	51.7
	5%	DB	51.8	52.3	62.6	71.7	79.1	82.4	84.4	83.5	79.4	70.4	61.2	53.6
		MCWB	47.9	45.4	51.7	57.3	64.7	69.7	71.8	70.4	67.2	60.7	52.8	49.3
	10%	DB	45.1	45.8	55.4	66.1	74.5	80.6	82.1	81.0	75.2	65.9	56.8	47.7
		MCWB	40.7	40.1	47.7	54.5	62.9	68.2	70.0	68.6	65.3	58.7	50.5	44.5
	0.4%	WB	57.0	56.5	60.3	64.9	72.0	75.2	77.8	75.9	74.3	70.4	60.9	57.5
		MCDB	61.1	61.5	71.2	76.4	82.3	84.5	87.5	85.0	83.6	76.1	66.7	61.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	53.4	51.5	56.7	61.5	69.0	73.4	75.4	73.8	71.4	66.2	57.3	53.2
		MCDB	56.0	55.2	64.7	71.4	78.4	83.0	84.5	81.8	79.5	71.9	62.4	55.9
	5%	WB	47.0	46.0	53.3	59.1	66.9	71.6	73.8	72.1	69.3	62.2	54.4	48.8
		MCDB	49.9	50.5	59.8	67.8	75.6	80.4	82.1	79.7	76.3	67.9	59.4	52.2
	10%	WB	40.9	40.7	48.8	56.5	64.5	69.8	72.1	70.7	67.3	59.1	51.1	43.9
		MCDB	44.8	46.3	56.5	64.7	72.1	77.2	79.4	77.6	73.1	65.5	57.0	47.5
(35) Mean Daily Temperature Range	5% DB	MDBR	14.4	16.1	18.1	21.4	21.0	19.9	20.0	19.7	19.9	18.7	15.8	13.3
		MCDBR	18.6	23.6	25.7	27.7	25.6	23.1	22.5	22.8	23.8	23.8	21.8	20.5
		MCWBR	15.9	18.3	16.1	15.7	13.3	11.5	10.4	10.6	12.1	14.3	15.5	17.0
	5% WB	MCDBR	17.6	21.4	21.9	22.2	21.4	19.9	18.6	18.4	19.4	19.6	18.8	18.2
		MCWBR	16.4	18.5	15.8	14.6	12.1	10.8	9.7	10.0	10.8	13.5	15.2	17.1
(40) Clear-Sky Solar Irradiance	taub		0.303	0.328	0.354	0.378	0.416	0.451	0.454	0.447	0.383	0.362	0.318	0.300
	taud		2.444	2.352	2.364	2.330	2.236	2.172	2.181	2.190	2.346	2.406	2.511	2.503
	Ebn at Noon		272	279	284	284	274	263	261	259	271	264	265	264
	Edh at Noon		25	32	35	39	44	47	46	44	35	30	23	22
	RadAvg		466	733	1041	1420	1658	1804	1822	1641	1334	880	590	385
(45) All-Sky Solar Radiation	RadStd		43	100	104	143	145	151	127	96	137	88	79	41

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (37 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+207	+97

Nomenclature: See separate page